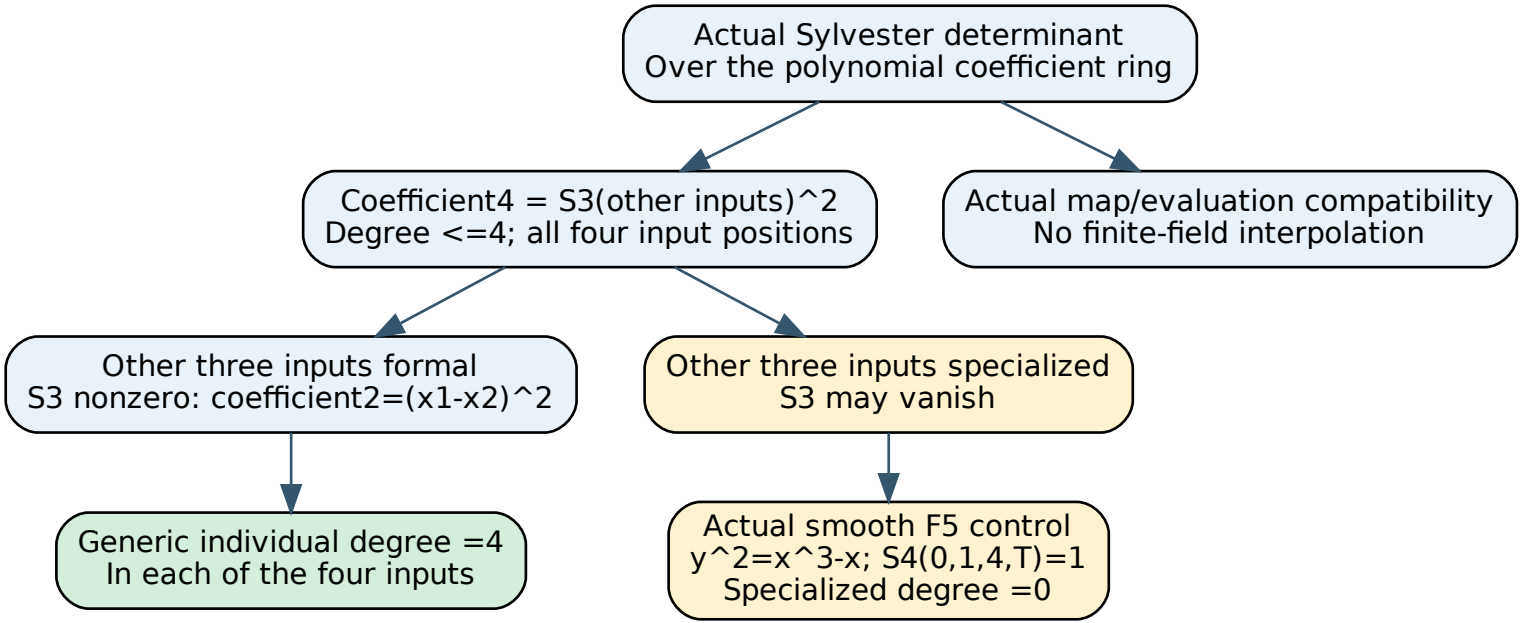
Actual Sylvester determinant
Over the polynomial coefficient ring
Coefficient4 = S3(other inputs)^2
Degree <=4; all four input positions
Actual map/evaluation compatibility
No finite-field interpolation
Other three inputs formal
S3 nonzero: coefficient2=(x1-x2)^2
Other three inputs specialized
S3 may vanish
Generic individual degree =4
In each of the four inputs
Actual smooth F5 control
y^2=x^3-x; S4(0,1,4,T)=1
Specialized degree =0